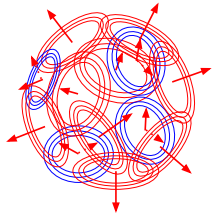
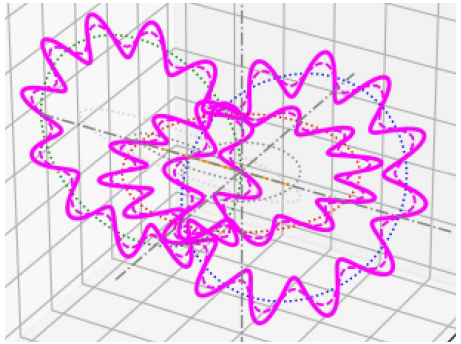
R-space Theory 1.0

- Geometric Integration of Relativity and Quantum Mechanics -



Chap.9:
What are Neutrinos?

Part 1 | Mechanism of the Weak Force



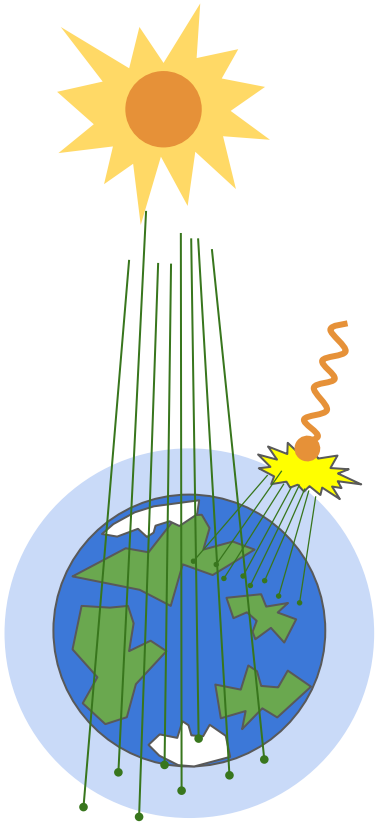
1. What are Neutrinos?

Neutrinos carry no electric charge, and because of this lack of charge, they do not interact via the electromagnetic force.

Furthermore, their mass is so minuscule that they are almost unaffected by gravity.

Consequently, they hardly interact with matter and possess the ability to pass through most substances, making them extremely difficult particles to detect.

However, neutrinos are ubiquitous in the universe, generated in diverse processes such as radioactive decay, nuclear fusion within the Sun, and collisions between cosmic rays and the atmosphere.



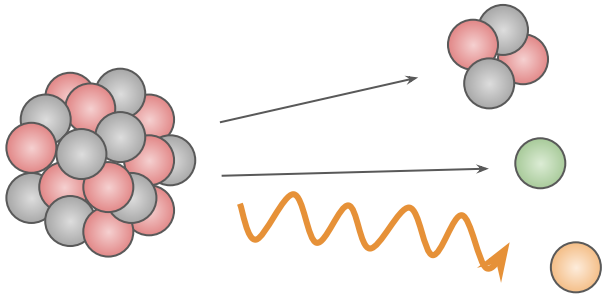
1. What are Neutrinos?

The existence of neutrinos was proposed by Wolfgang Pauli in 1930.

At that time, three types of radiation were known: alpha, beta, and gamma rays.

Today, we understand that alpha rays are helium nuclei, beta rays are electrons, and gamma rays are high-energy photons (electromagnetic waves).

In beta decay, an atom A decays into atom B while emitting a beta ray (electron).



α ray; helium nuclei

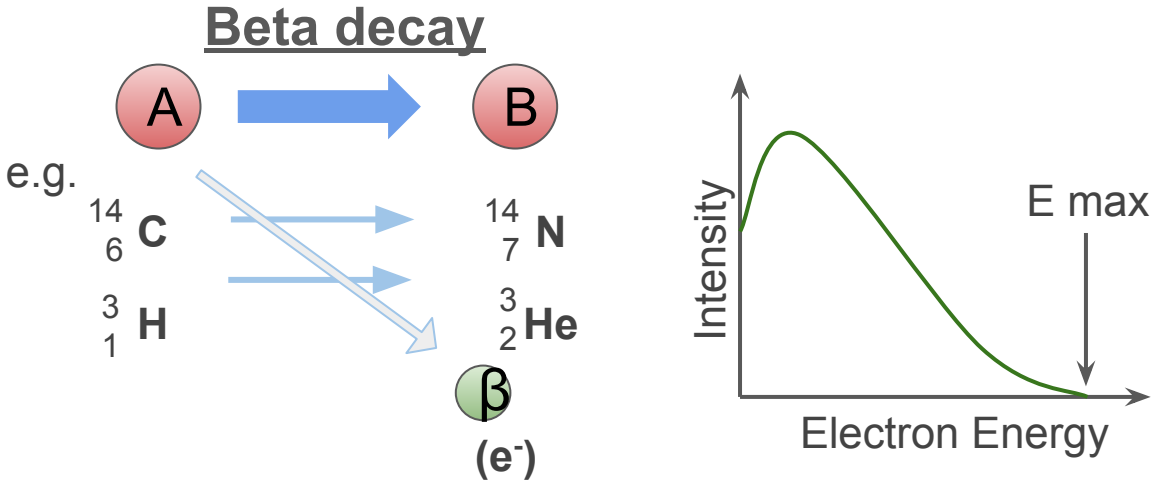
β ray; electrons

γ ray; high-energy photons

1. What are Neutrinos?

The energy of the beta ray was initially expected to be discrete (constant), reflecting the energy difference between atom A and atom B.

However, experiments revealed that the energy spectrum of electrons in beta decay is continuous, which appeared to violate the "Law of Conservation of Energy".

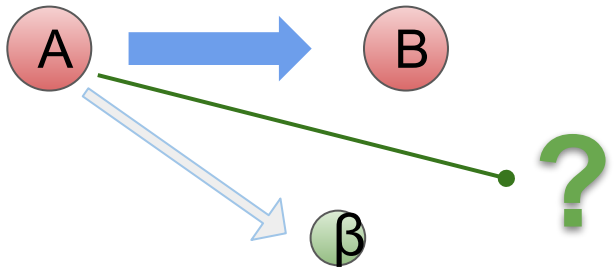


1. What are Neutrinos?

The conservation of energy is a fundamental cornerstone of physics.

To preserve this law, Pauli proposed a "desperate remedy" (Verzweiflungsausweg), hypothesizing the existence of a neutral, extremely light, and weakly interacting particle that carries away the missing energy.

Twenty-six years later, in 1956, Clyde Cowan and Frederick Reines successfully confirmed the existence of neutrinos by detecting electron antineutrinos from a nuclear reactor via inverse beta decay (anti-ν_e + p $\rightarrow$ e^+ + n).

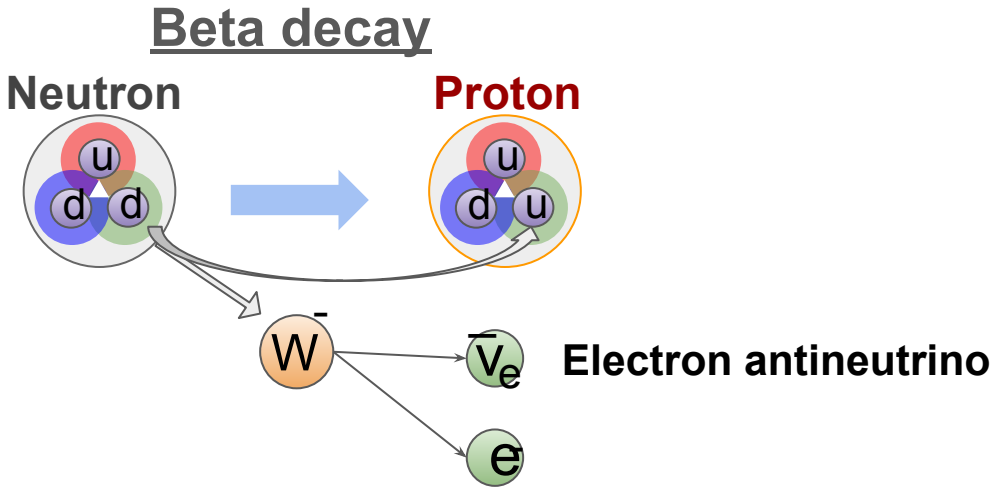


A neutral, extremely light, and weakly interacting particle

1. What are Neutrinos?

In the current Standard Model, beta decay is described as a process where a down quark within a neutron decays into an up quark by emitting a W^- boson, which mediates the weak interaction.

This change in quark flavor transforms the neutron into a proton. Subsequently, the W^- boson decays into an electron and an electron antineutrino.



1. What are Neutrinos?

Additionally, there are three generations of neutrinos—electron neutrino (ν_e), muon neutrino (ν_μ), and tau neutrino (ν_τ)—corresponding to the three generations of quarks and leptons in the Standard Model.

These neutrinos can transform into one another, a phenomenon known as neutrino oscillation.



Neutrino oscillation

1. What are Neutrinos?

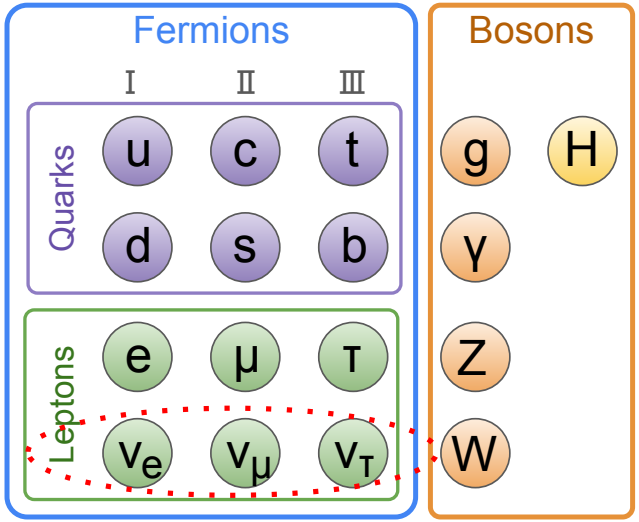
Quarks

	Mass	Charge	Spin
Up-type	≈2.16 MeV/c2～172.57 Gev/c2	+2/3	1/2
Down-type	≈4.7 MeV/c2～4.183 GeV/c2	-1/3	1/2

Leptons

	Mass	Charge	Spin
Electron family	≈0.511 MeV/c2～11.77693 Gev/c2	-1	1/2
Neutrino family	<0.8 eV/c2～18.2 MeV/c2	0	1/2

Standard Model of particle physics



1. What are Neutrinos?

Due to the extreme difficulty of observation, neutrinos remain mysterious.

While the discovery of neutrino oscillation implies that neutrinos have non-zero mass, research continues into whether one of the three types might remain massless.

Furthermore, the weak interaction involving neutrinos exhibits unique phenomena such as parity violation, and there are ongoing theories suggesting neutrinos may be Majorana particles, meaning they are their own antiparticles.

2. Weak Interaction and Neutrino Model

The weak interaction is one of the four fundamental forces of nature (electromagnetic force, strong force, weak force, and gravity).

It is an extremely short-range, low-intensity force caused by the exchange of "very heavy gauge particles" called W and Z bosons.

The weak force is the only force capable of changing the type of particle, such as changing a quark's flavor from d to u; as mentioned, beta decay is also caused by the weak force.



W Boson
≃80.376 GeV/c2



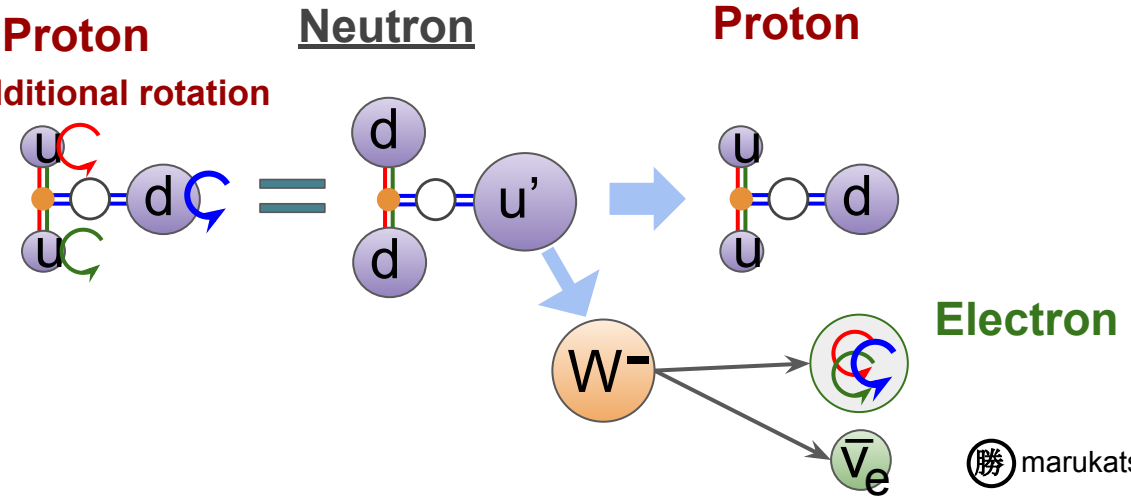
Z Boson
≃91.188 GeV/c2

2. Weak Interaction and Neutrino Model

In R-space Theory, as explained in Chapters 7 and 8, beta decay occurs when the three quarks composing a neutron release their rotational energy as a W^- boson and return to a proton state.

The rotational energy released as a W^- boson then forms an electron. The excess energy at that time is the electron antineutrino.

Beta decay in R-space Theory

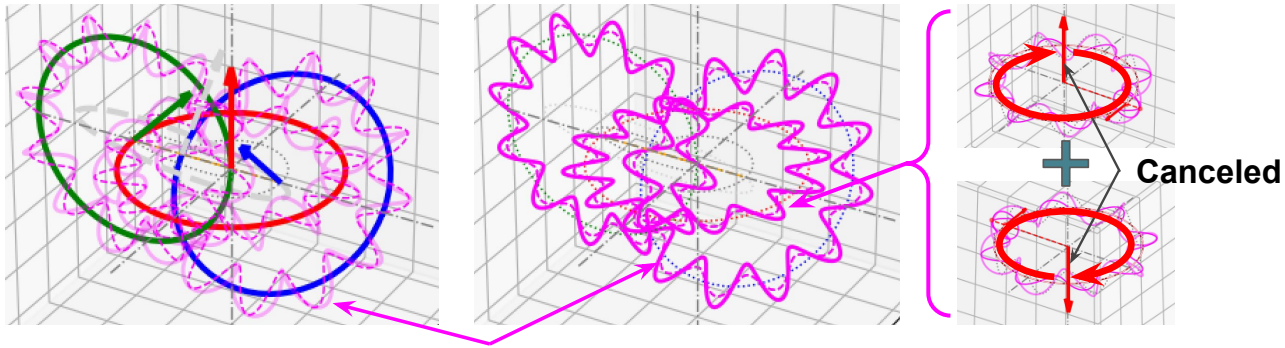


2. Weak Interaction and Neutrino Model

Since neutrinos have no charge and very small (or zero) mass, it is thought that the primary rotations of their rotation vectors cancel each other out.

In this state, the vibrations of the color charges superpose to form a standing wave, and the neutrino is considered to be the particle-like manifestation of this vibration due to color confinement.

Electron model **Neutrino model**



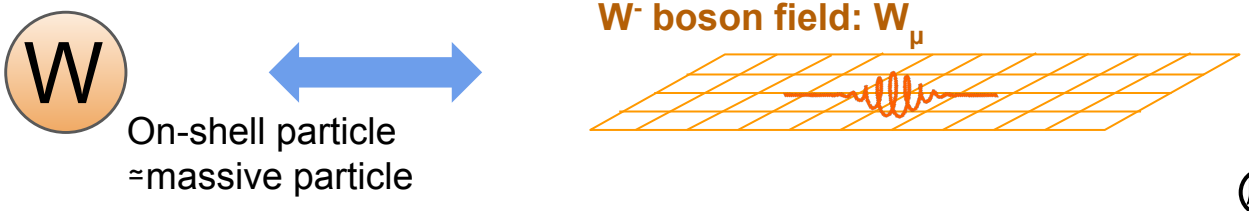
Color charge rotation / vibration

3. Standard Model W Bosons and Gauge Fields

In the Standard Model, the weak interaction in beta decay is mediated by the W^- boson.

The W^- boson is an exceptionally heavy gauge boson with a mass of approximately 80 GeV/c^2, and was discovered in 1983 through the UA1 and UA2 experiments at CERN.

It is important to note that while the W^- boson has been observed as a real particle, the W^- boson involved in beta decay acts as a virtual particle—an excited state of the W^- boson field W_μ—rather than a real, on-shell particle.



3. Standard Model W Bosons and Gauge Fields

The Standard Model is built upon the framework of Quantum Field Theory (QFT).

In this framework, a "field" is an intrinsic property of spacetime, and elementary particles manifest as "excitations (vibration modes)" of these fields.

Fields are broadly categorized into matter fields and gauge fields.

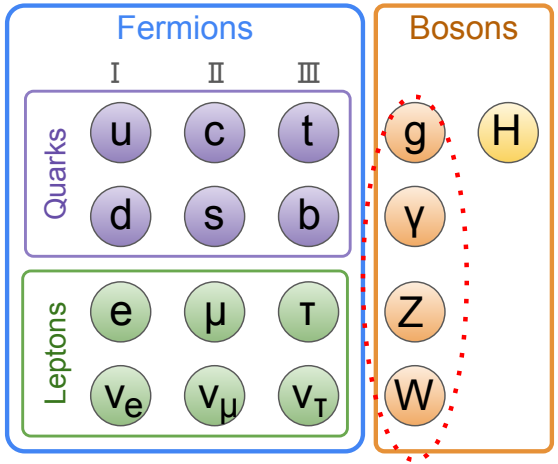
Matter fields include the electron field and quark fields; their excitations appear as the particles (fermions) that constitute matter.

Conversely, fields associated with force-carrying particles (gauge bosons) such as photons, W/Z bosons, and gluons are known as gauge fields.

3. Standard Model W Bosons and Gauge Fields

Interaction	Gauge field	Gauge particle	Mass	Charge	Spin
Electromagnetic interaction	A_μ	Photon (γ)	0	0	1
Weak interaction	W_μ^a ($a = 1,2,3$)	Z Boson	≈91.188 GeV/c2	0	1
		W Boson	≈80.376 GeV/c2	±1	1
Strong interaction	G_μ^a ($a = 1 \sim 8$)	Gluon (g)	0	0	1
Gravity	($h_{\mu\nu}$ or $\Gamma_{\mu\nu}^{\rho}$)	Graviton (undiscovered)	0	0	2

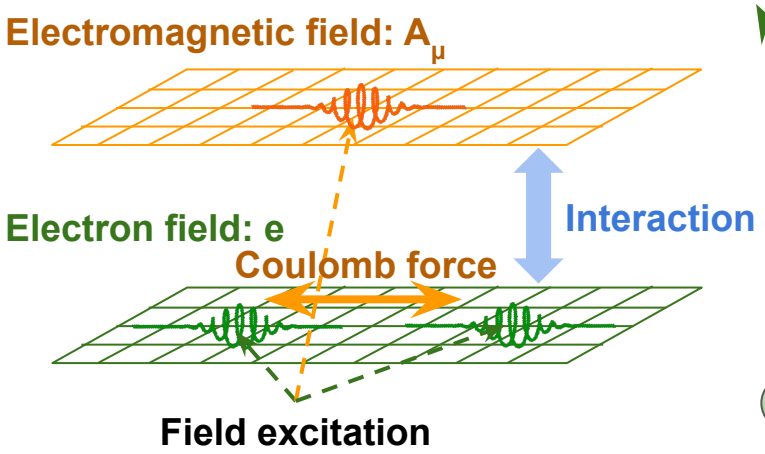
Standard Model of particle physics

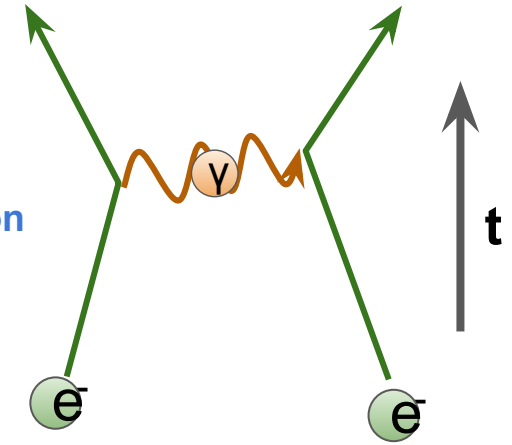


3. Standard Model W Bosons and Gauge Fields

Gauge fields emerge through local gauge symmetry and are coupled to matter fields, allowing them to mediate interactions.

For instance, the photon emerges from U(1) symmetry (specifically $U(1)_{em}$) as an excitation of the electromagnetic field (A_μ), interacting with charged particles like electrons or quarks to manifest as the Coulomb force.





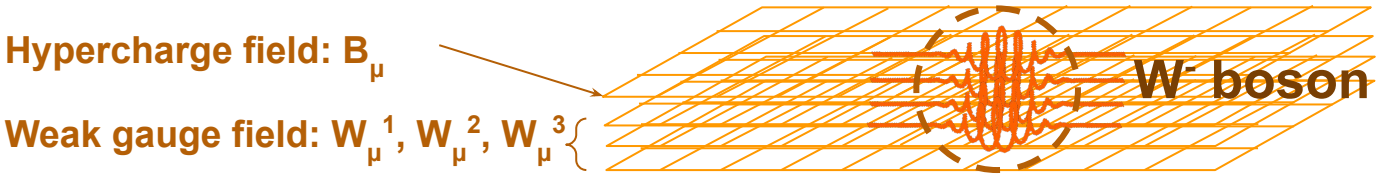
3. Standard Model W Bosons and Gauge Fields

Fermions, which constitute matter (quarks and leptons), possess U(1) and SU(2) symmetries (specifically $U(1)_Y$ hypercharge and $SU(2)_L$ weak isospin) .

Quarks, which participate in the strong interaction, also possess SU(3) color symmetry. The $U(1)_Y$ and $SU(2)_L$ symmetries give rise to four gauge fields: the B_μ field and the three $W_\mu{}^a$ fields ($W_\mu{}^1$, $W_\mu{}^2$, $W_\mu{}^3$) .

The W and Z bosons are excitations resulting from combinations of these fields.

Notably, the electromagnetic field A_μ is a mixing of the B_μ and $W_\mu{}^a$ fields.

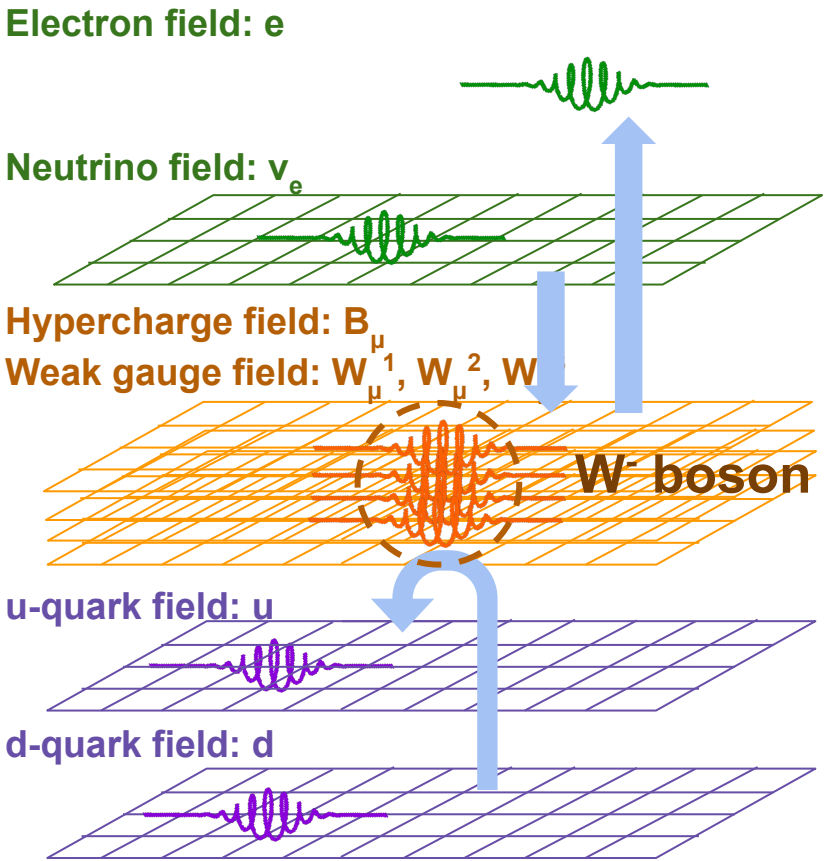


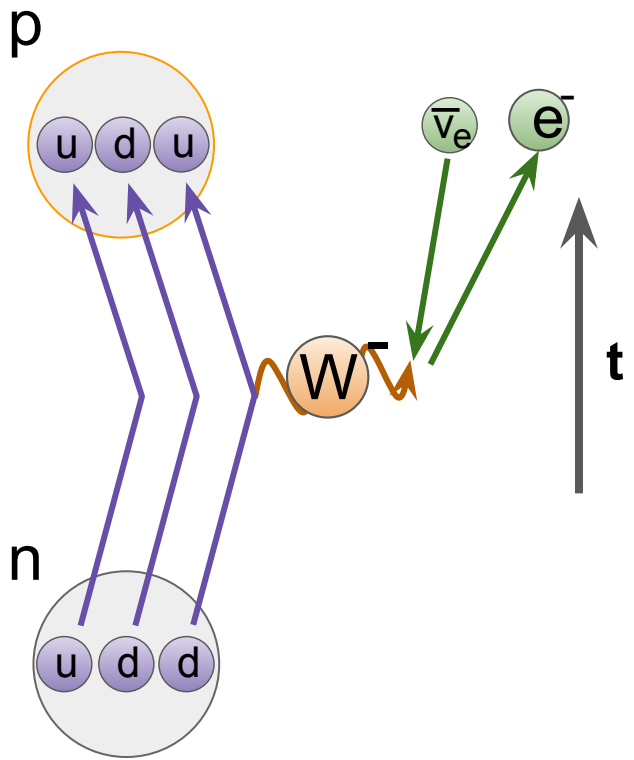
3. Standard Model W Bosons and Gauge Fields

The W_{μ}^{a} fields interact with matter fields, effectively "rotating" the internal excitation modes (the direction of internal vibration) of particles, thereby changing their identity (flavor).

In the process of beta decay, the excitations of the down quark, up quark, electron, and neutrino fields all interact via the excitation of the W_{μ} field.

3. Standard Model W Bosons and Gauge Fields





4. Fields in R-space Theory

In R-space Theory, all particles are represented by three rotation vectors (x, y, z) and their combinations.

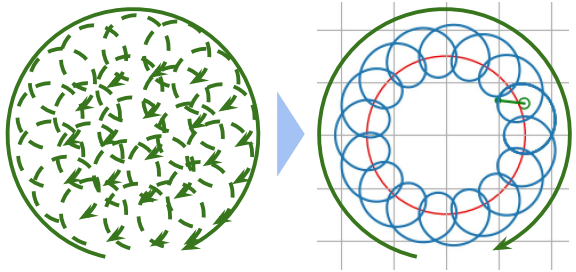
As explained in Chapter 4, these three rotation vectors are projected onto spacetime using each other's rotation as phase.

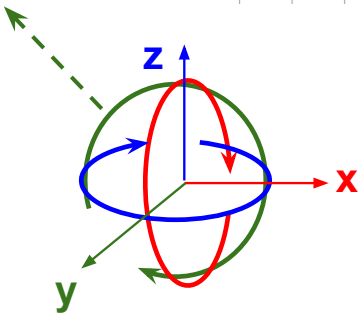
Rotation vectors also represent angular momentum and can be thought of as a collection of small vortices.

These small vortices are projected according to their respective phases, and their superposition corresponds to the "field in R-space Theory".

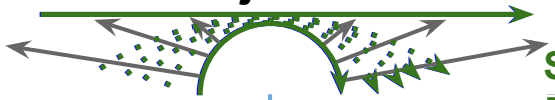
4. Fields in R-space Theory

Rotation vector (collection of small angular-momentum vortices)



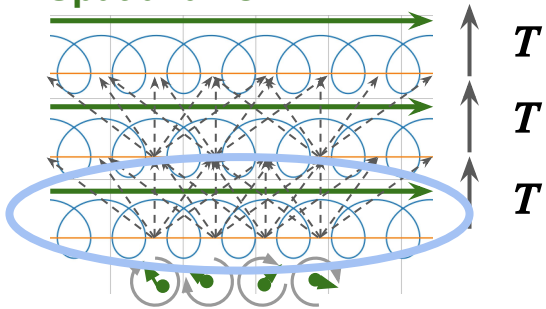


Projection



Spatial axis

Time

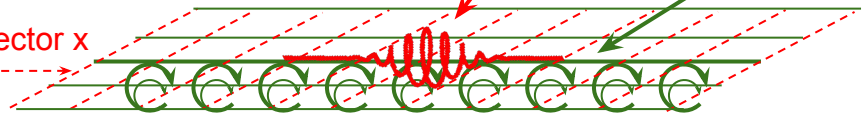


Projected rotation vector x

Projection by rotation vector y

Fields in R-space Theory

Projection by rotation vector x



4. Fields in R-space Theory

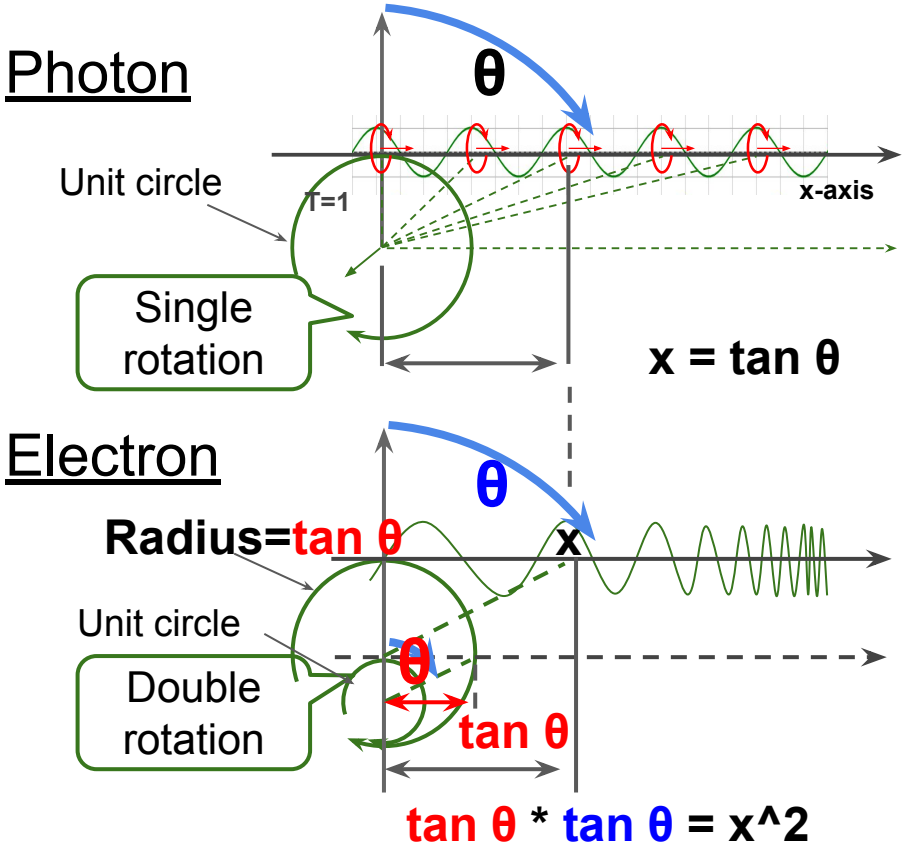
There are two types of this phase rotation: "single rotation" for photons and "double rotation" for fermions like electrons.

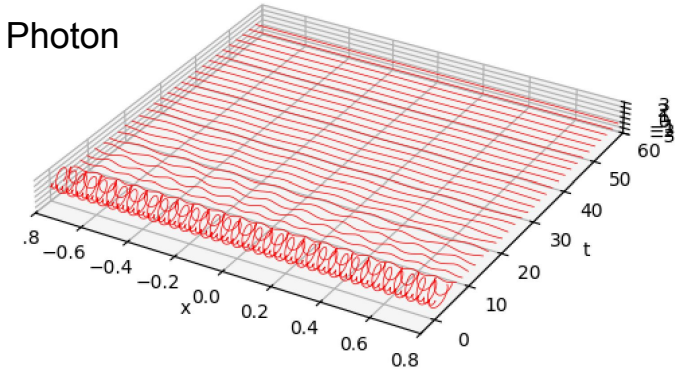
Particles corresponding to gauge particles (gauge fields) must be single-rotational.

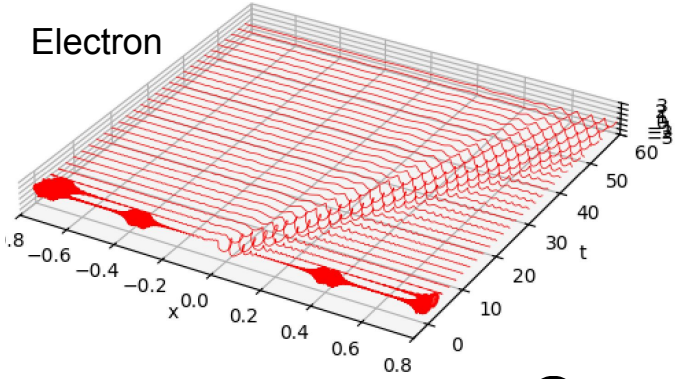
This is because, as explained in Chapter 6, in the case of single rotation, the phase spreads (is projected) in proportion to the first power of distance, allowing the phase to spread evenly across spacetime (while it spreads and overlaps probabilistically, the result maintains the original spatial phase relationship).

Conversely, in double rotation, the phase spreads in proportion to the square of the distance, causing spatial phase relationships to mix and disperse.

4. Fields in R-space Theory







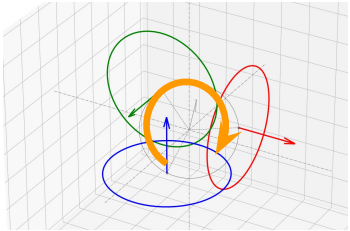
4. Fields in R-space Theory

To transmit interaction to particles (matter fields) as a gauge field, the field value (phase) must be determined locally.

Therefore, gauge fields (gauge particles; bosons) in R-space Theory are single-rotational. (Note: Previously, colors were assigned by dimension (x, y, z), but henceforth, colors are assigned by particle/field for clarity).

Gauge field (photon)

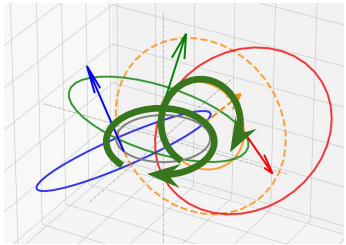
Single rotation





Matter field (electron)

Double rotation





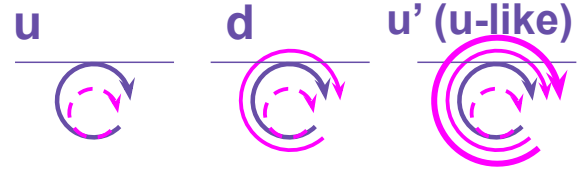
4. Fields in R-space Theory

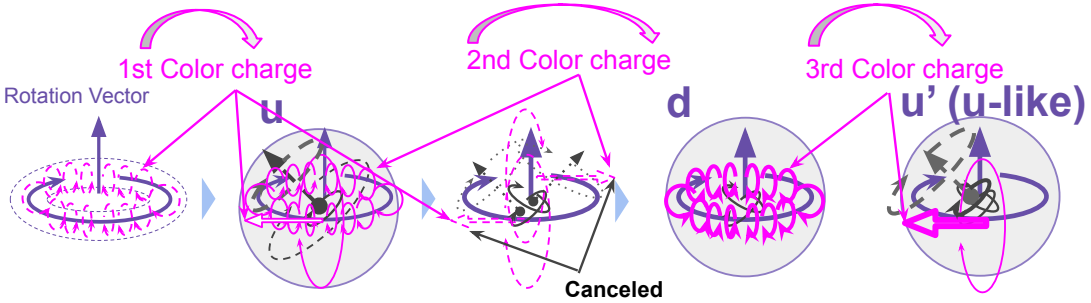
Matter fields (fermions) are double-rotational, but in quarks, a color charge rotation is added for each flavor, accompanied by a gluon field arising from the vibration of the color charge.

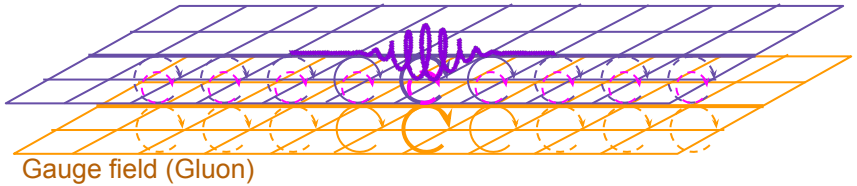
The type of rotation corresponds to "spin": single rotation corresponds to spin 1, and double rotation corresponds to spin 1/2.

Matter field (quark)

Double rotation + color charge rotation with gluon field







5. W Boson Field in R-space Theory

Beta decay as a field in R-space Theory is as follows: the three quarks composing a neutron (d, d, u-like) are in an unstable state; their rotation state changes, releasing rotational energy and returning to the proton state (u, u, d).

Since these three rotational energies are released from the double-rotation state of the quarks, they are in a single-rotation state, i.e., spin 1.

In R-space Theory, as explained in Chapter 6, the double-rotation state is the factor that generates mass.

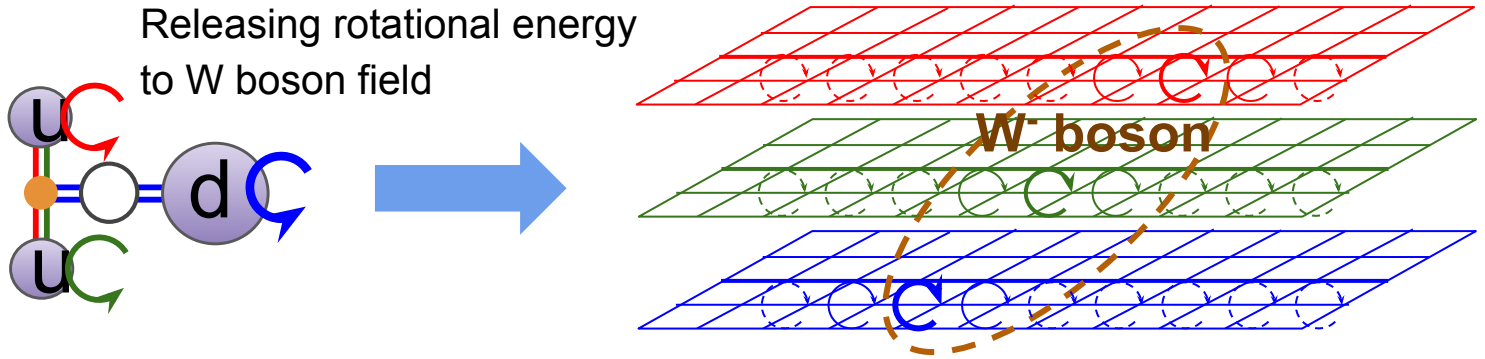
Therefore, in this single-rotation state, the mass is zero.

5. W Boson Field in R-space Theory

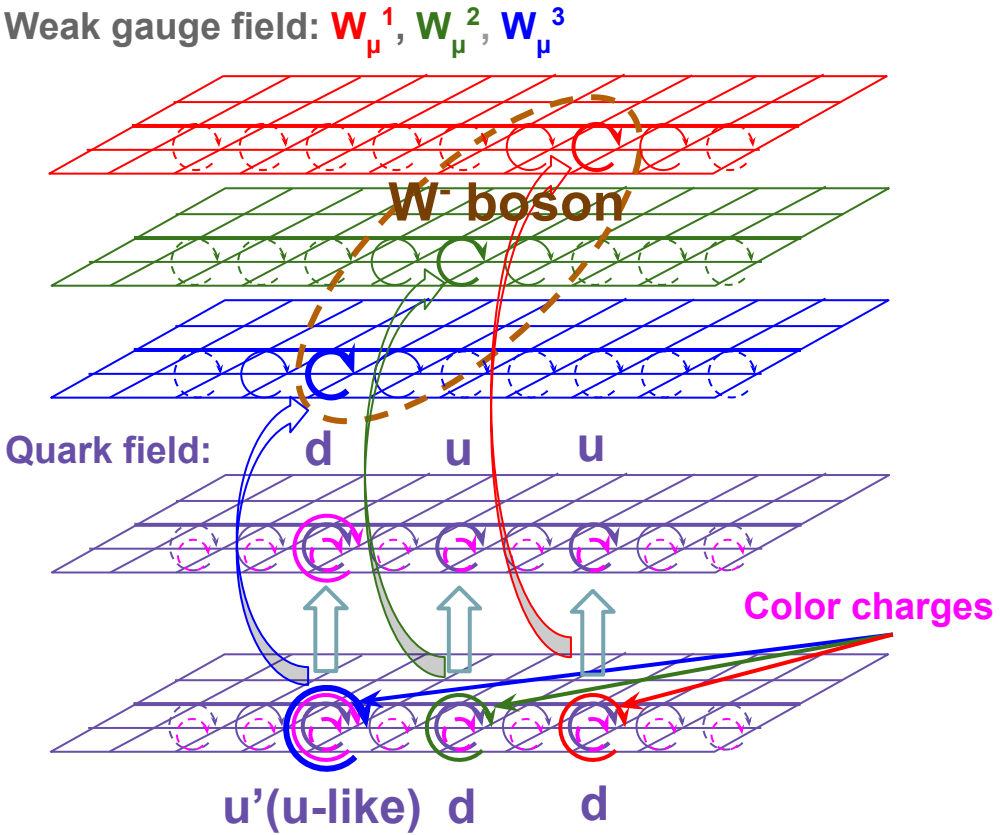
This state is the virtual particle state of the W boson—the W_μ field state.

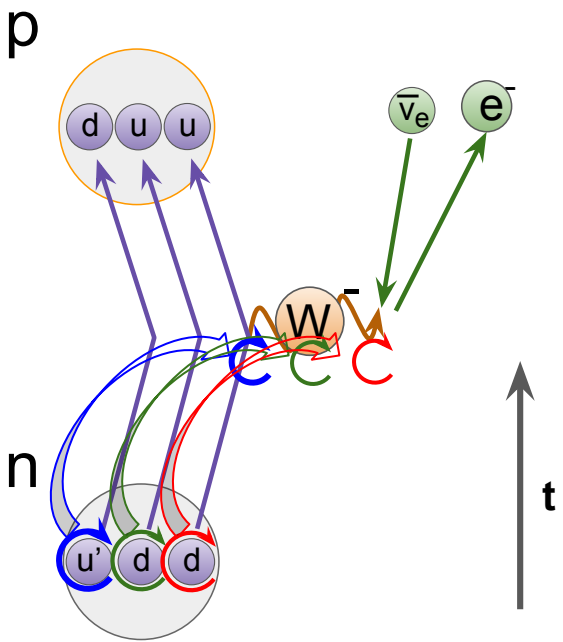
The phase fields of these three rotation vectors correspond to the three W_μ^a fields (W_μ^1, W_μ^2, W_μ^3).

Note that in R-space Theory, since the rotation vectors themselves carry charge, there is nothing corresponding to the B_μ field.



5. W Boson Field in R-space Theory



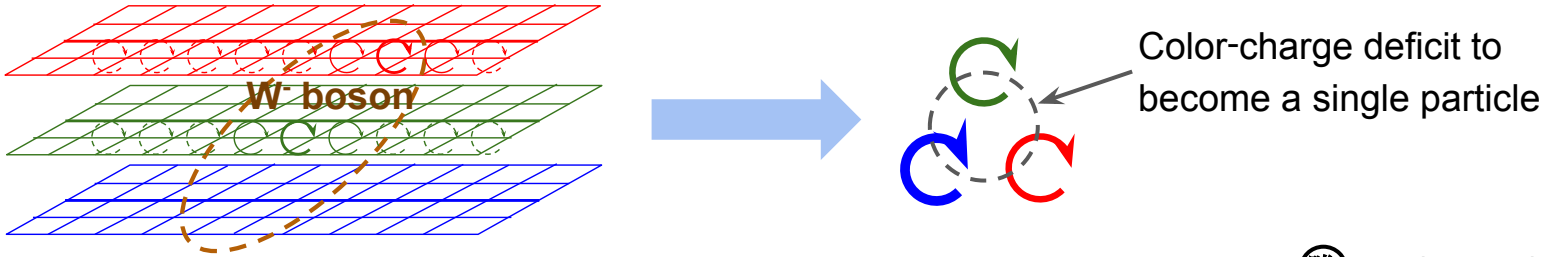


5. W Boson Field in R-space Theory

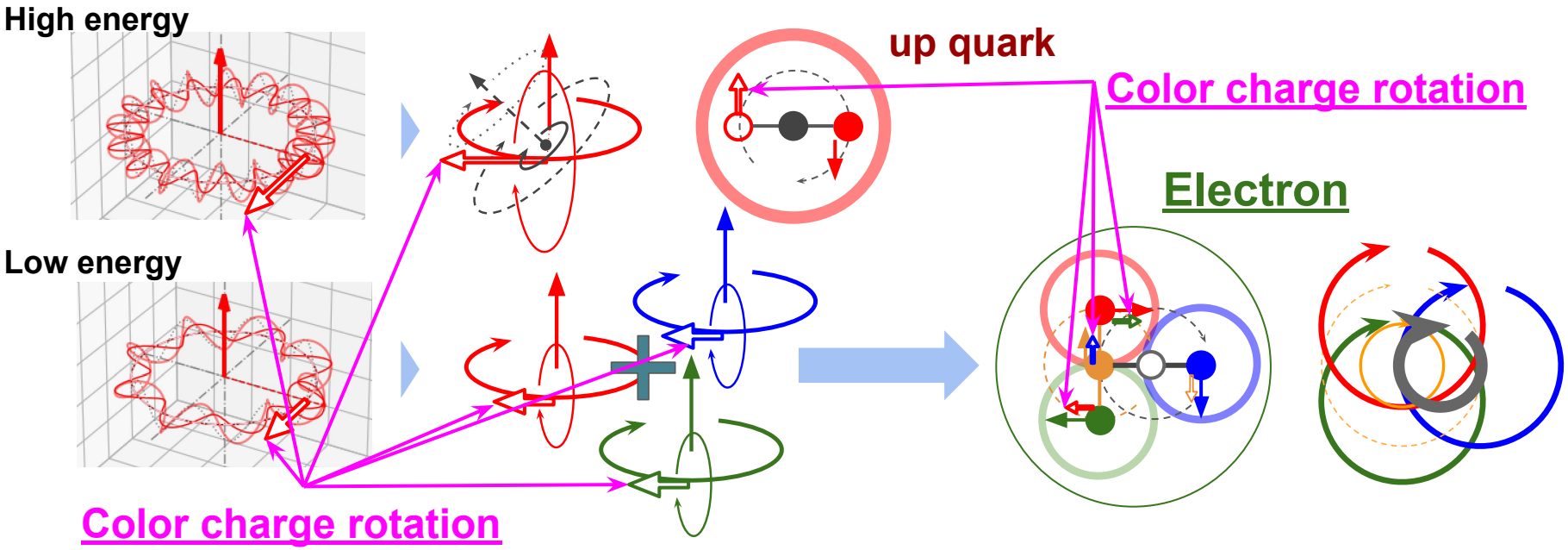
Just as quarks cannot exist in isolation, these three rotation vectors (W_μ fields) cannot exist independently.

Therefore, they immediately attempt to become a single particle (an excitation of the electron field).

As explained in Chapter 7, what confines the three rotation vectors composing an electron is the low-energy rotation of color charges (vibration of the angular momentum of the rotation vectors).



5. W Boson Field in R-space Theory



5. W Boson Field in R-space Theory

To compensate for the color charge lacking for color confinement, the three rotation vectors of the W_{μ}^{a} field use part of their energy to undergo "pair production" of color charges.

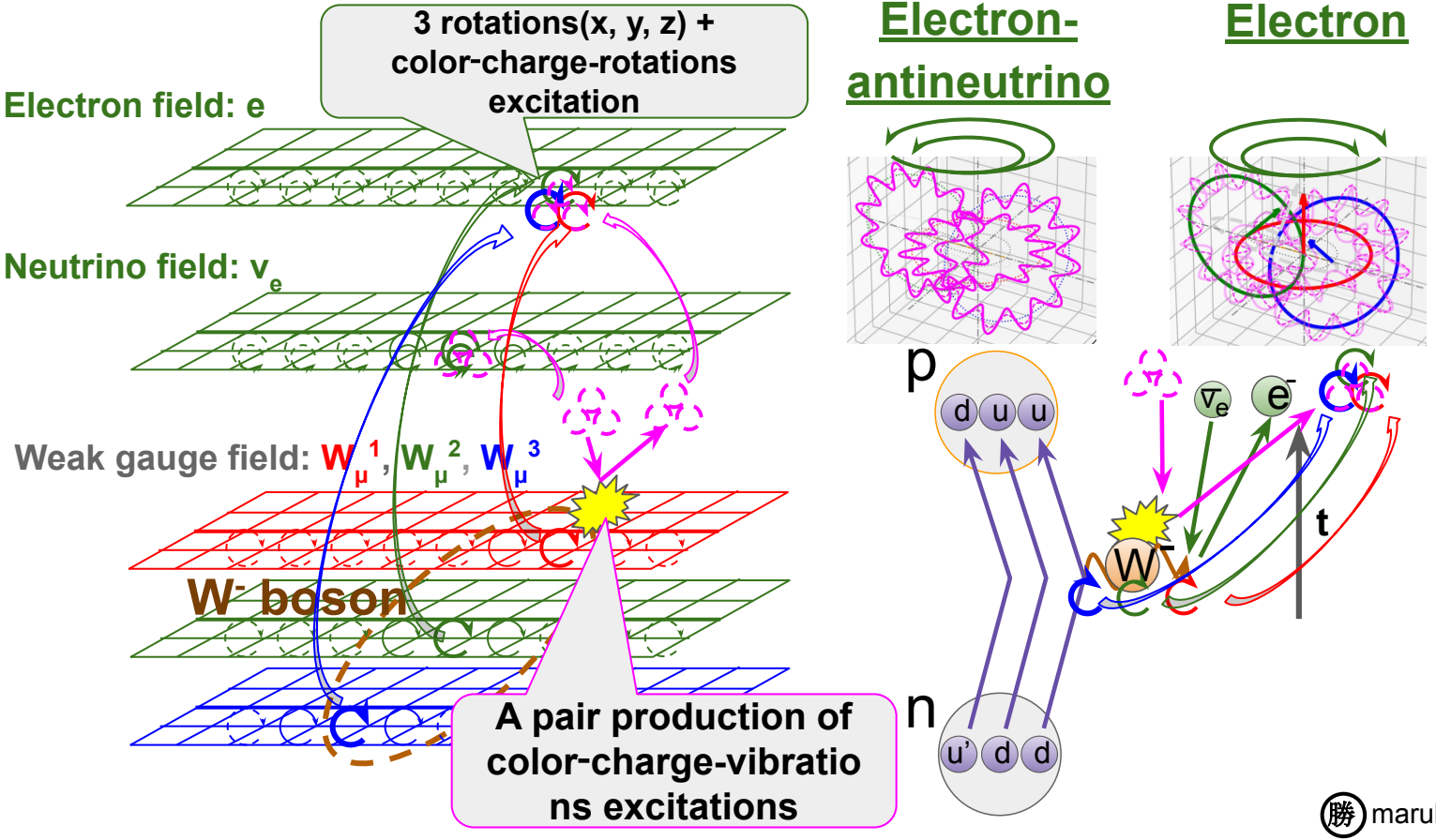
Using one of the pair-produced color charges, the three rotation vectors of the W_{μ}^{a} field begin double rotation to become an electron.

The other is emitted as a neutrino.

At that time, to conserve angular momentum (spin), it emerges as an electron antineutrino (an antiparticle) with a spin opposite to that of the electron.

However, as explained in Section 2, because the rotation vectors of the neutrino are canceled out, only the direction of revolution(orbital motion) is reversed.

5. W Boson Field in R-space Theory

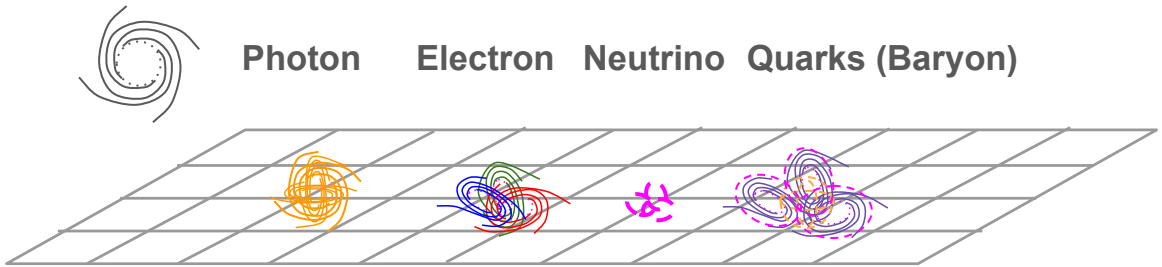


5. W Boson Field in R-space Theory

The above is the mechanism of beta decay as a field in R-space Theory.

As mentioned, in R-space Theory, both gauge fields and matter fields are fields of the same angular momentum; they are a single angular momentum field differing only in rotation mode.

It is like vortices on a water surface approaching, separating, or colliding to change shape.



5. W Boson Field in R-space Theory

However, although they are called vortices, they appear as such as a result of R-space phase rotations being projected and superposed in spacetime.

Parity symmetry violation in weak interactions and neutrino oscillation will be explained in subsequent chapters.

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This is all for my presentation.

Thanks for watching!

